Data Analytics Project 4: Data Visualization

DecodeLabs Industrial Training Kit | Dataset-based visualization and business insights

Objective

Create clear visual representations of the provided e-commerce order dataset, select appropriate charts, and communicate actionable insights. The project follows the guide: match chart type to the question, maintain honest axes, reduce chart clutter, and use conclusion-led storytelling.

Dataset Overview

Metric	Result
Total Orders	1200
Unique Customers	1189
Number of Products	7
Total Units Sold	3535
Total Sales	1264761.96
Average Order Value	1053.97
Highest Order Value	3456.4
Lowest Order Value	11.39
Exact Duplicate Rows	0
Missing Coupon Codes	309
Data Period	2023-01-01 to 2025-06-30

Key Findings

• Total recorded sales are 1,264,761.96 across 1,200 orders and 1,189 unique customers.

• Chair is the highest-sales product, generating 195,620.11 across 178 orders.

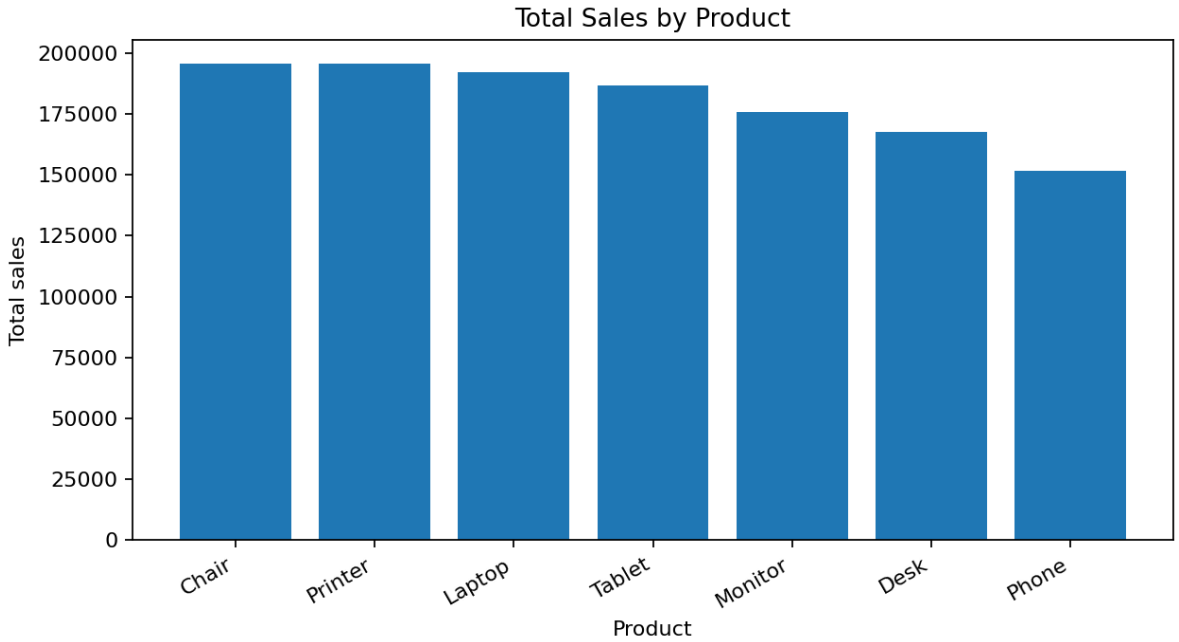
• Cancelled is the most frequent order status with 250 orders.

• Instagram is the top referral source by sales, generating 275,285.45.

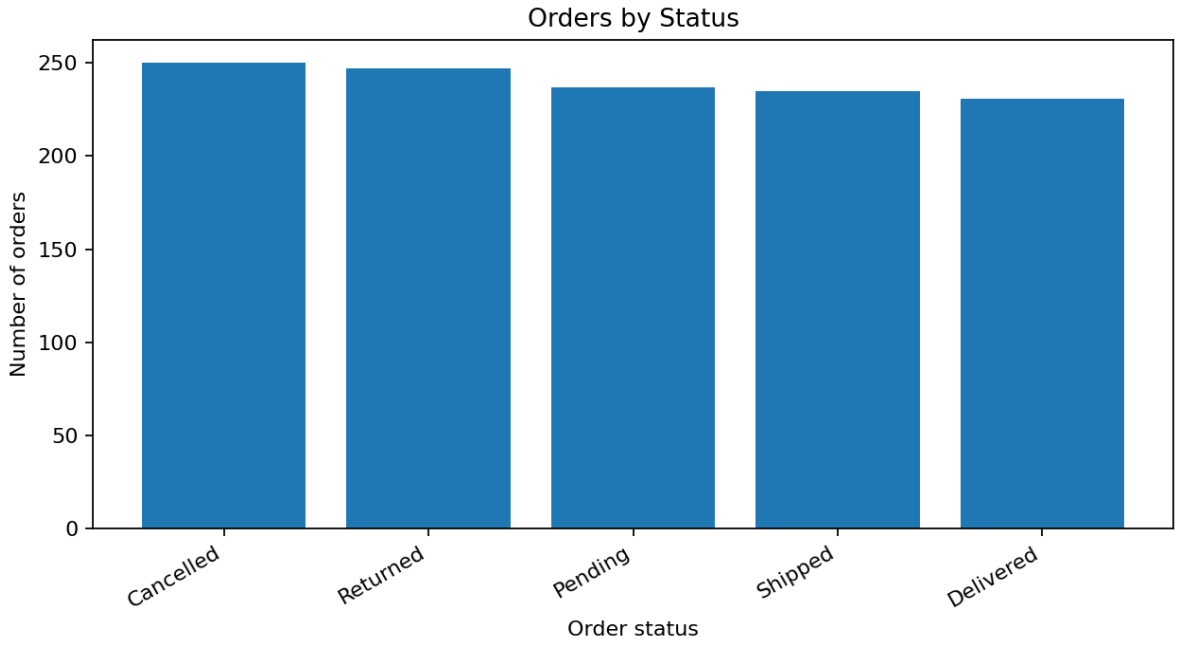
• The dataset contains 0 exact duplicate rows and 309 missing coupon codes. Missing coupon values are treated as 'No coupon' only in coupon analysis.

Visual Analysis

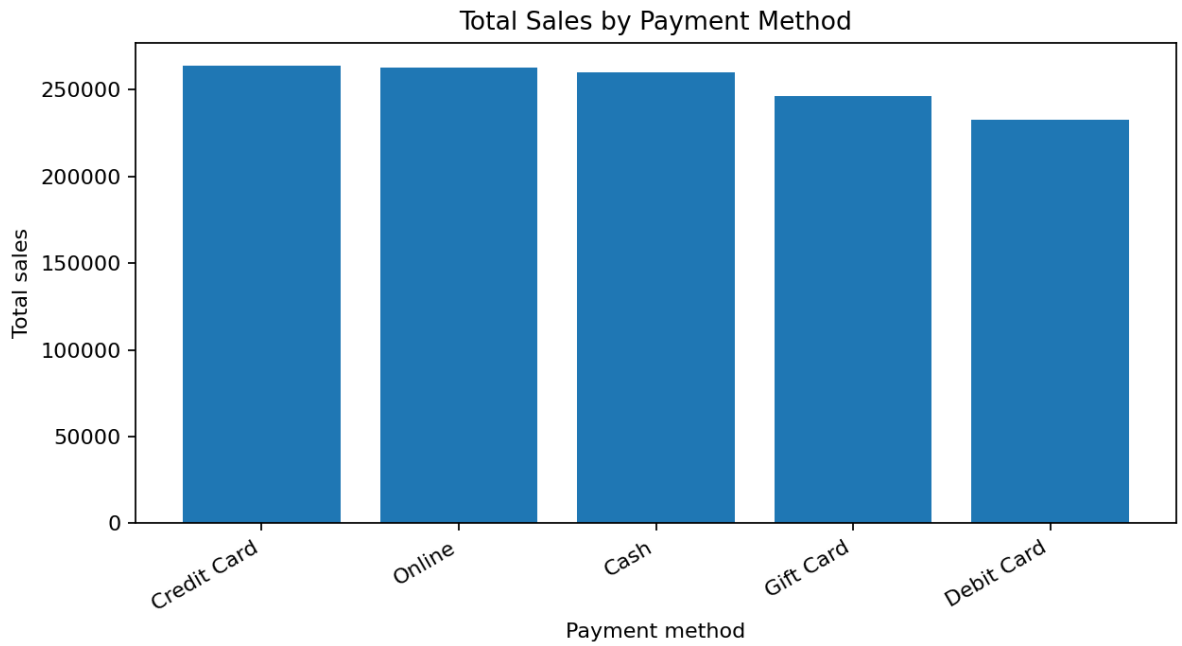
1. Total Sales by Product



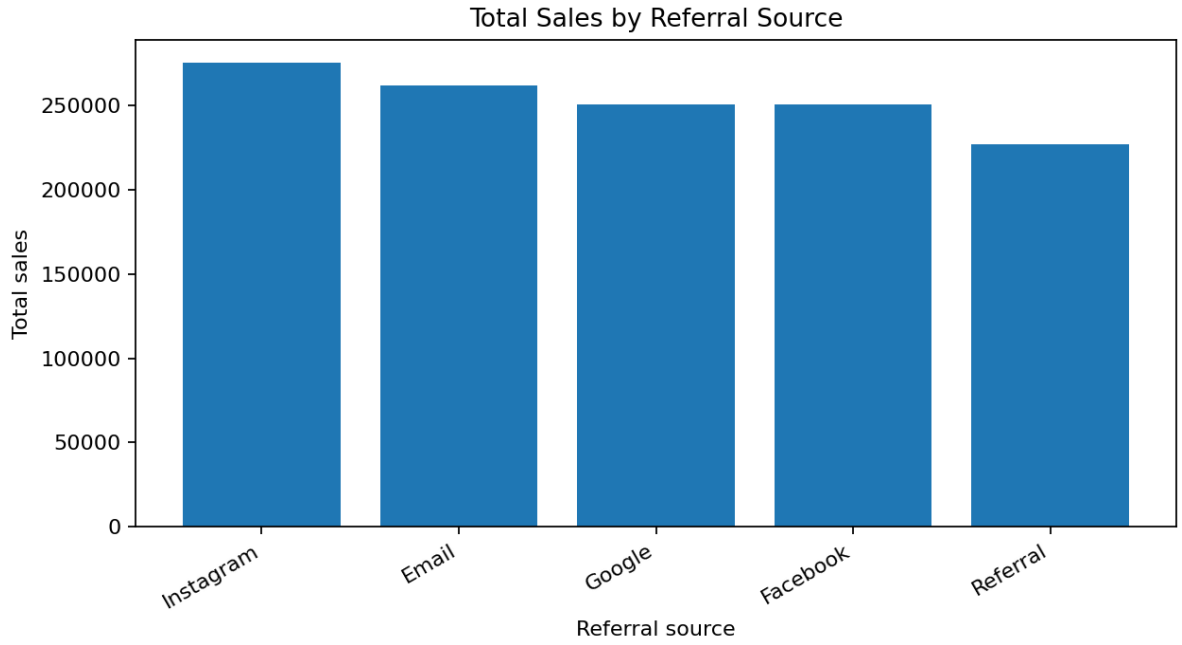
2. Orders by Status



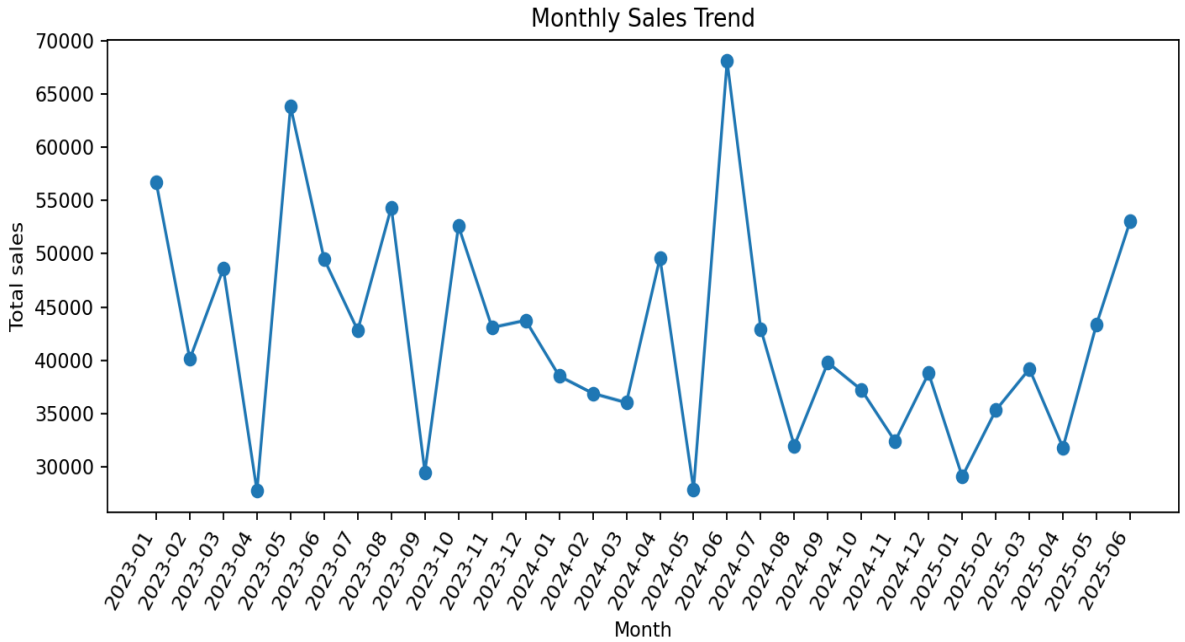
3. Sales by Payment Method



4. Sales by Referral Source



5. Monthly Sales Trend



Recommendations

1. Review product-level sales and stock allocation, prioritizing products with the strongest total sales while checking profit margins before making inventory decisions.

2. Investigate cancelled and pending orders to identify avoidable order failures and improve fulfilment.

3. Compare referral sources using both order count and sales value; focus marketing spend on sources that perform well on business outcomes, not just traffic.

4. Track monthly sales over time and annotate unusually high or low months before presenting results to management.

5. Improve data quality by validating coupon codes and checking date, quantity, and price fields during data entry.

Chart design notes: bar charts compare categories; a line chart shows monthly trend. Axes are labelled, chart titles describe the metric, and no 3D effects are used. Values are based on the supplied TotalPrice field; currency is not specified in the dataset.